

Prasun Majumder

B.Tech. – Electronics and Communication Engineering
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https://scholar.google.com/citations?user=2D45pH0AAAAJ&hl=en
https://github.com/prasyin3636

Professional Profile

Electronics and Communication Engineering undergraduate with interdisciplinary experience in embedded systems, IoT, AI and ML, sensor development, sustainable materials, waste valorization and environmental monitoring. Interested in Research and Development, intelligent healthcare, energy storage, bio-derived materials and AI-enabled sensing, with a focus on innovative hardware prototyping and sustainable technology.

Education

B.Tech. in Electronics and Communication Engineering 2024 – Present Techno International New Town, Kolkata, 700156

- YGPA: 7.14 (1st Year), 6.75 (2nd Year).
- CGPA: 6.95 (up to 2nd Year).

Higher Secondary (Class XII) **2023**
Delhi World Public School, Kalyani, Nadia, 741235 *Percentage: 71.5%*

Secondary (Class X) **2021**
Springdale High School, Kalyani, Nadia, 741235 *Percentage: 86%*

Research and Academic Experience

Research and Internship Experience

Summer Research Internship Programme 2026 – IEEE SMC **May 15 – Aug. 15, 2026**
IEEE Systems, Man, and Cybernetics Society Student Branch Chapter, Kalyani Government Engineering College Research Intern

- Successfully completed the Summer Research Internship Programme 2026 organized by the IEEE Systems, Man and Cybernetics Society Student Branch Chapter, Kalyani Government Engineering College, in collaboration with the IEEE Systems, Man, and Cybernetics Society Chapter, IEEE Kolkata Section.
- The topic was on **Circular Buffer-Based Feature Extraction and Edge AI for Early Pressure Ulcer Risk Prediction in Bedridden Patients**.
- Explored circular-buffer-based approaches for efficient physiological signal processing and feature extraction for continuous monitoring applications.
- Investigated Edge AI approaches for early risk prediction and intelligent healthcare monitoring in bedridden patients.
- Focused on computationally efficient and deployable healthcare intelligence suitable for resource-constrained edge devices.

IEEE EMBS National Student Internship Program 2026 **Jun. 1 – Jun. 30, 2026**
IEEE EMBS Pune Chapter *Research Intern*

- Successfully completed a one-month internship under the IEEE EMBS Pune Chapter Online Student Internship Program 2026.
- Worked on the project **Dementia-IOT Based Solution**, focusing on the application of IoT and intelligent sensing technologies for dementia-related healthcare.
- Explored IoT-enabled approaches for healthcare monitoring and technology-assisted support for patients and caregivers.
- Applied knowledge of embedded systems, sensors, data acquisition and intelligent monitoring to a healthcare-oriented research problem.

IoT Intern **Sep. 2025 – Nov. 2025**
1Stop.ai *Intern*

- Developed and implemented three end-to-end IoT prototypes using ESP32, Python and cloud-based platforms including ThingSpeak and Blynk.
- Designed a smart factory simulation integrating sensors, ESP32-based embedded control and IoT connectivity for real-time monitoring.
- Developed a remote health monitoring prototype for collecting and transmitting physiological parameters through an IoT-enabled system.
- Built a live weather-data monitoring system using IoT sensors, ESP32, Python and cloud-based data visualization.
- Worked on sensor interfacing, embedded programming, wireless communication, cloud integration and real-time IoT data visualization.

IEEE EMBS Student Internship Program 2025 **Jun. 15 – Jul. 15, 2025**
IEEE EMBS Pune Chapter *Research Intern*

- Designed and prototyped an AI-assisted wearable healthcare system for monitoring physiological and motion-related signals associated with stress, seizure, tremor and cardiac anomalies.
- Integrated ECG, heart-rate, temperature, SpO_2, galvanic skin response and motion sensing using MAX30100, AD8232, LM35, GSR and MPU6050 with ESP32.
- Implemented threshold-based alerts, OLED visualization, adaptive scoring and baseline calibration.
- Simulated sensor behaviour using Wokwi and Tinkercad prior to hardware implementation.
- Focused on affordable and scalable healthcare technology for rural and underserved populations.

Patents and Intellectual Property

- **Published Indian Patent Application – IN202631096003 A1:** *An Integrated Continuous Microcontroller-Controlled Parthenium Biorefinery System and Process for Safe Harvesting, Fractionation of Cellulose, Hemicellulose and Lignin, and Production of Bioethanol, Pyrolytic Oil and Biochar.* Listed as an inventor.

Major Research and Innovation Projects

NEERSETU – Smart Water Purification and Quality Monitoring System
Developing a smart purification and monitoring platform for rural and mining-affected areas, with particular emphasis on water treatment, contaminant removal and low-cost sensing. The system incorporates pyrolytic-char-based adsorption, chitosan coagulation, ion-exchange/polishing media, UV-C treatment and optical water-quality monitoring.

- Developed multilayer filtration architecture using pyrolytic char, activated pyrolytic char, silica, sodium alginate, calcium chloride, banana-peel-derived carbonaceous material, zeolite and chitosan-based polishing components.
- Integrated pH, TDS/EC, turbidity and temperature sensing with BME280, BH1750 and MLX90614 for environmental monitoring.
- Investigated filtration performance using laboratory-prepared muddy, soil-contaminated, ash-containing and tap-water samples.
- Performed preliminary image analysis using ImageJ for estimation of clear-water recovery and particulate settling behaviour.
- Explored UV-C chamber integration for post-filtration disinfection and anthocyanin-based pH indication.

ECOCELLUPAK – Parthenium Biomass Valorization
Sustainable conversion of Parthenium hysterophorus biomass into value-added products including bio-alcohol, lignin-rich fractions, bio-charcoal and activated carbon.

- Investigated dilute sulfuric-acid pre-hydrolysis, alkaline delignification, neutralization, fermentation and distillation routes for biomass valorization.
- Obtained cellulose, hemicellulose, lignin and bio-charcoal fractions from dried Parthenium biomass.
- Developed activated carbon through pyrolysis followed by chemical activation using KOH.
- Investigated the material using UV–Vis spectroscopy and electrical conductivity measurements.

FuelTure – Plastic Waste to Fuel
Developed a pyrolysis-based concept for converting HDPE and LDPE plastic waste into liquid fuel and other carbonaceous products.

- Developed and prototyped a small-scale plastic pyrolysis system.
- Investigated process heating, condensation and collection of pyrolysis products.
- Developed a pilot concept with an approximate prototype capacity of 20 L/day.

Waste-to-Sensor Conversion of Biomass Process Liquors
Development of an organic chemi-resistive humidity sensor using residual biomass-processing liquors.

- Investigated Parthenium biomass-processing liquors as a source of functional organic sensing materials.
- Developed an organic sensing layer containing humic- and phenolic-like constituents for humidity detection.
- Fabricated chemi-resistive humidity-sensing structures using pencil-graphite electrodes.
- Investigated resistance variations under different humidity conditions, associated with moisture-induced ionic and protonic conduction.
- Characterized the sensing material using SEM and FTIR analysis to examine surface morphology and chemical functional groups.

Plastic-Derived Pyrolytic Charcoal-Based Artistic Ink
Research and development of a sustainable inorganic pigment ink using plastic-derived pyrolytic charcoal.

- Explored plastic-derived pyrolytic charcoal as a carbon-based pigment for sustainable artistic ink formulation.
- Developed a water-based ink formulation incorporating pyrolytic charcoal as the primary colourant.
- Investigated pigment dispersion, colour intensity, surface coverage and coating uniformity.
- Studied the potential of waste-derived carbon pigments as an alternative to conventional synthetic colourants.
- Focused on waste valorization, material sustainability and potential applications in artistic and functional coatings.

Parthenium Root-Derived Organic Fountain Pen Ink
Development of a sustainable brown-coloured organic ink using Parthenium root-derived carbon and bio-based formulation additives.

- Explored Parthenium root-derived carbon as a biomass-based pigment for eco-friendly brown-coloured ink.
- Formulated an aqueous ink incorporating tannin and xanthan gum to investigate colour development, dispersion and viscosity control.
- Investigated pigment stability, flow characteristics and writing performance for potential fountain pen applications.
- Evaluated the potential of plant-derived materials as sustainable alternatives to conventional ink ingredients.
- Focused on biomass valorization, bio-based material development and environmentally conscious ink formulation.

HDPE and LDPE Plastic Waste-Derived Activated Pyrolytic Char
Development of activated carbonaceous material from HDPE and LDPE plastic waste using a pyrolysis and chemical activation system with a 5 kg processing capacity.

- Investigated HDPE and LDPE plastic waste as feedstocks for producing pyrolytic char through thermal decomposition.
- Worked with a pyrolysis machinery setup designed to process up to 5 kg of plastic waste per batch for char production and subsequent activation.
- Developed activated pyrolytic char through chemical activation to improve surface properties and potential adsorption performance.
- Explored plastic-derived carbon materials for potential applications in filtration, environmental remediation, and functional composites.
- Focused on plastic waste valorization, resource recovery, and the development of value-added carbonaceous materials.

Engineering and IoT Projects

- **SmartDrip – TinyML-Based Intelligent Saline Monitoring:** Developed an ESP32-based system using a 940 nm IR LED, TEFT4300 phototransistor, ADS1115 16-bit ADC and BH1750 ambient-light sensor. Implemented Haar wavelet denoising, adaptive thresholding, flow-feature extraction and an embedded Random Forest model converted using Micromlgen to classify normal, low, blocked, fast and empty saline conditions. Integrated OLED display, buzzer alerts, battery monitoring and IoT data logging.
- **Guardian Helm – AI-Powered Smart Safety Helmet:** Integrated MPU6050 motion sensing and ESP32-CAM with a YOLOv5-based lightweight computer-vision model trained using pothole images sourced from IEEE DataPort. Designed a road-hazard detection concept combining visual inference with motion sensing to support rider safety and potential real-time alerts.
- **NeuroSense – Multimodal Wearable Health Monitoring:** Designed a wearable monitoring concept integrating MAX30102, MPU6050 and physiological sensing for heart rate, SpO_2, motion, tremor and stress-related indicators. Explored sensor fusion and intelligent pattern recognition for identifying abnormal physiological and movement patterns.
- **MedAssistBot – Human-Centric Healthcare Robot:** Designed an assistive robotic concept for elderly and vulnerable patients integrating ESP32-based health monitoring, MAX30102, temperature sensing, MQ-2 gas detection, flame sensing, medicine reminders and voice alerts. Proposed RSSI-based indoor navigation and a Streamlit interface with generative AI API integration for educational assistance, Wikipedia-based information retrieval, summarization and AI-powered problem-solving.
- **12-V Battery Health Monitoring for E-Rickshaws:** Developed a battery-monitoring prototype concept using voltage-divider networks, resistors, filtering capacitors and sensor interfacing for voltage acquisition and embedded analysis. Explored current measurement, temperature monitoring, battery-status estimation and threshold-based alerts for preventive maintenance.
- **NeerSetu – Smart Water Purification and Optical Quality Monitoring:** Designed a multi-stage filtration system using plastic-derived pyrolytic char, activated carbon, silica, sodium alginate, $CaCl_2$, banana-peel carbon and an alginate–banana peel–zeolite–chitosan polishing composite. Integrated pH, TDS/EC, turbidity and temperature sensing with optical

measurements using white, red, green, blue, IR and UV LEDs, BPW21R and TEFT4300 photodetectors, and LM358 signal conditioning. Explored optical attenuation, multispectral response analysis, reagent-assisted testing, UV disinfection and sensor-based treatment decisions with recirculation for further treatment when quality criteria are not met.

- **ECOCELLUPAK_T16 – Parthenium Biorefinery and Waste Valorization:** Investigated Parthenium hysterophorus fractionation using dilute H_2SO_4 hydrolysis, NaOH delignification, filtration, pH adjustment and distillation to recover cellulose, hemicellulose-rich fractions, lignin and bio-alcohol. Explored lignin pyrolysis for bio-oil, activated-carbon production and conversion of residual process liquors into organic chemi-resistive humidity sensors. Contributed to a waste-to-value approach integrating chemical engineering, sustainable materials and functional sensing.
- **NeuroCare – Dementia Risk Assessment Using Edge AI:** Developed a multimodal monitoring framework using ESP32, MAX30102, MPU6050, LM35 and INMP441 to analyse physiological signals, movement, falls and sound-related behavioural changes. Designed a Random Forest classifier in Python/Google Colab and converted it into embedded C++ using Micromlgen for local risk classification and caregiver alerts.
- **Epilepsy Monitoring System:** Explored wearable seizure-event monitoring using time-windowed sensor data, embedded processing and machine-learning-based classification. Designed an edge-monitoring approach for identifying abnormal movement patterns and generating alerts for caregiver awareness.
- **Pressure-Ulcer Risk Prediction Using Edge AI:** Designed a wearable risk-monitoring framework for bedridden patients using MPU6050 motion data at 50 Hz, MAX30102 heart rate and SpO_2, and temperature sensing. Implemented circular-buffer-based windowing, statistical and physiological feature extraction, and an 11-feature Random Forest model converted with Micromlgen for ESP32 inference. Proposed four risk categories and wireless caregiver alerts without requiring expensive pressure mats or smart beds.
- **AI-Assisted Diabetic Foot Ulcer Assessment:** Explored an image-based ulcer assessment system using computer vision and AI to analyse wound images and support identification of ulcer-related visual characteristics. Focused on accessible preliminary assessment and potential referral support rather than replacing clinical diagnosis.
- **Automatic Fire-Extinguishing Robot Car:** Developed a sensor-guided robotic prototype integrating flame detection, Arduino-based control, motor actuation and a fire-extinguishing mechanism. Explored directional flame localization and autonomous approach-and-response logic to reduce manual intervention in small-scale fire scenarios.
- **Automatic Parking System:** Built an Arduino-based parking prototype using ultrasonic distance measurement and servo-motor actuation. Implemented obstacle-distance thresholding and automated barrier control to demonstrate contactless vehicle detection and controlled parking access.
- **Obstacle-Avoiding Autonomous Car:** Developed an Arduino Uno-based vehicle using ultrasonic ranging and motor-driver control. Implemented distance-based navigation logic with forward movement, obstacle detection, stopping and directional turning to improve autonomous obstacle response.
- **Automatic Street Lighting:** Developed an Arduino Nano-based automatic lighting prototype using IR sensing and conditional control logic. Explored occupancy-responsive switching to reduce unnecessary illumination and power consumption compared with continuously operated lighting.
- **Smart Home Fire Alarm:** Designed a flame-sensor-based fire detection system with microcontroller-driven buzzer alerts. Implemented threshold-based event detection and rapid local warning for residential fire-safety applications.
- **Arduino-Based Radar System:** Developed a servo-scanned ultrasonic sensing prototype to measure obstacle distance across angular positions. Combined sensor readings with Arduino processing and graphical visualization to represent nearby objects in a radar-like display.
- **PID-Based Line-Following Robot:** Developed an autonomous vehicle using line sensors, motor drivers and PID control. Implemented error-based steering correction to reduce path deviation, improve tracking stability and produce smoother movement along a predefined route.
- **Python Snake Game with MySQL (Class XII):** Programmed an interactive Snake game in Python with keyboard controls, collision detection, score management and game-state logic. Integrated MySQL to store and retrieve player scores, demonstrating foundational programming and database connectivity.
- **Ticket Booking Interface Clone:** Developed a responsive booking-interface prototype using HTML, CSS and JavaScript, incorporating interactive seat or selection controls and QR-code generation to demonstrate digital ticketing workflows.
- **FuelTure Student Startup Website:** Developed a web interface for a student startup focused on converting HDPE and LDPE plastic waste into fuel through pyrolysis. Presented the project concept, process workflow, sustainability objectives and prototype information using HTML, CSS and JavaScript.
- **Automatic Train Braking – Conceptual Safety System:** Designed a conceptual embedded railway-safety architecture incorporating sensor-based abnormal-condition and potential derailment detection, decision logic and automatic braking signals. Focused on fault response, timely intervention and fail-safe design considerations.
- **2D/3D Engineering Design and CAD Modelling:** Created and explored two-dimensional technical drawings and three-dimensional structural models for engineering prototypes, filtration assemblies, robotic mechanisms and system layouts using CAD tools. Focused on component placement, assembly visualization, dimensional representation and practical prototype planning.

Technical Skills

Programming	C, Python, Embedded C, HTML, CSS, JavaScript, SQL
AI/ML Models	Random Forest, Decision Trees, TinyML, YOLOv5, Computer Vision, Edge AI, Sensor Fusion, Feature Engineering, Classification
Signal Processing	Haar Wavelet Transform, Discrete Wavelet Transform (DWT), Fourier Transform (FFT), Filtering, Noise Reduction, Adaptive Thresholding, Statistical Feature Extraction
Embedded Systems	Arduino Uno/Nano, ESP32, ESP8266, Arduino IDE, GPIO, I^2C, ADC, PWM, IoT Prototyping
Sensors	MAX30100/02/05, AD8232, MPU6050, MLX90614, LM35, DHT22, BH1750, LDR, IR, Flame, Ultrasonic, pH, TDS/EC, Turbidity, Optical Sensors
Electronics & Hardware	MOSFETs, BJTs, Resistors, Capacitors, Diodes, Relays, Servo/DC Motors, BMS, Buck/Boost Converters, Voltage Dividers, Pumps, Heating Systems
AI & Development Tools	Google Colab, PyTorch, TensorFlow Lite Micro, Streamlit, FastAPI, LangChain, Azure OpenAI, ThingSpeak, Firebase
Chemistry & Materials	Biomass Fractionation, Acid Hydrolysis, Alkali Delignification, Pyrolysis, Chemical Activation, Adsorption, pH Analysis, FTIR, SEM, XRD, UV-Vis Spectroscopy
Physics & Instrumentation	Optical Sensing, Photodetection, Electrical Resistance, Conductivity, Electromagnetic Energy Harvesting, Thermodynamics, Heat Transfer, Sensor Calibration
Mathematics & Analytics	Statistics, Probability, Linear Algebra, Numerical Analysis, Data Visualization, Signal Analysis, Regression, Model Evaluation
Research & Simulation Tools	MATLAB (Basics), ImageJ, Wokwi, Tinkercad, AutoCAD, KiCad (Learning), Microsoft Office Suite
Research Domains	Sustainable Materials, Plastic Waste Valorization, Parthenium Biorefinery, Activated Carbon, Energy Storage, Environmental Monitoring, Biomedical Wearables, Intelligent Healthcare, AI-Enabled Sensing

Publications, Conferences and Presentations

Selected Publications

- **Majumder, P., Chakraborty, P.P.** "Double Resource Recovery from Plastic Waste: Development and Characterisation of an Eco-Friendly Carbon-Based Ink from Plastic Pyrolysis-Derived Carbon and Recycled Tea Leaf ..." *International Journal of Research and Innovation in Applied Science*, 2026.
- **Majumder, P.** "Edge-AI Enabled Affordable Wearable for Intelligent Monitoring of Physiological Patterns Associated with Neuro-Cognitive Disorders." *International Journal of Research and Innovation in Applied Science*, 2026.
- **Majumder, P., Chakraborty, A., Saha, M., Sharma, A., Patra, R., Das, A.** "Transformation of Plastic Waste into Pyrolytic Carbon Via Pyrolysis: Structural Examination by X-Ray Diffraction Analysis for Sustainable Environmental Applications." *International Journal of Research and Innovation in Applied Science*, 2026.
- **Dey, S., Majumder, P., Barik, S.** "A Federated Learning Framework for Collaborative Ageing and State of Health Prediction in E-Rickshaws." *2025 International Conference on Computing, Intelligence, and Application (CIACON)*, 2025.
- **Majumder, P.** "Trackless Autonomous Trams: A Catalyst for Sustainable Innovation and Technology." *International Conference on Management and Business Practices (ICMBP-2025)*, 2025.

Conference Presentations and Innovation Showcases

- **ICMBP-2025, Aliah University:** Presented research on **"Trackless Autonomous Trams: A Catalyst for Sustainable Innovation and Technology."**
- **CIACON 2025, Dr. B.C. Roy Engineering College:** Presented **"A Federated Learning Framework for Collaborative Ageing and State of Health Prediction in E-Rickshaws"** (IEEE Conference Record ID: 65473).
- **International Conference on Sustainable Development, SXUK:** Presented **"Green Horizons,"** focusing on sustainable technological innovation and development.
- **Regional Science Congress 2026, University of Kalyani:** Delivered an oral presentation on **Parthenium Biomass Valorization** and sustainable waste-to-value conversion.
- **Emerging Trends in Chemical Sciences Towards Sustainability & Interdisciplinarity (ETCSTSI-2026), Bhowanipore:** Presented **"Integrated Fractionation and Valorization of *Parthenium hysterophorus* for Sustainable Biofuel and Bio-product Generation: An Experimental Process Study."**
- **RDAMSE 2026, Surtech College:** Presented research on the development and application of a **humidity-sensing layer** for resistive/environmental sensing.
- **IESIA 2026, IEM:** Alongside a team member, presented **"TinyML with ESP32-Based Embedded Sensing for an Intelligent AI-Powered Real-Time Saline Drip Monitoring and Alert System."** The work integrated ESP32-based embedded sensing, optical drip detection, Haar wavelet signal processing, feature extraction and TinyML-based classification for real-time saline-flow monitoring.
- **OPTIMA Tech Fest 2025, IIT Kharagpur:** Presented **FuelTure**, a sustainable plastic-waste-to-fuel project based on pyrolysis, as part of the innovation/industrial design showcase.
- **Incubes 2025, Techno International New Town:** Presented **FuelTure**, a sustainable plastic-waste valorization and fuel-generation project.
- **MCA Inter-College Poster Presentation Competitions, Techno International New Town:** Participated in and

presented research/innovation posters in two inter-college poster presentation competitions.

- **3rd Edition of IDE Bootcamp – Phase I, Dr. B.C. Roy Engineering College:** Presented the **Parthenium Biomass Valorization** project, focusing on sustainable biomass fractionation and conversion into biofuels and value-added bio-products.

Research and Academic Activities

- Conducted research involving sustainable materials, plastic-waste valorisation, pyrolytic carbon, functional carbon materials, AI-enabled wearable healthcare systems, federated learning and intelligent sensing.
- Performed manuscript revision and reviewer-response preparation for research involving SEM and FTIR characterization.
- Engaged in scientific writing, conference presentation, research communication and interdisciplinary technology development.

Competitions, Hackathons and Innovation Challenges

- **Smart India Hackathon (SIH) 2025 – Hardware Edition:** Selected as a national finalist with team *ECOCELLU-PAK_T16*; secured **3rd place** in the college-level internal hardware hackathon.
- **Smart India Hackathon (SIH) 2026 – Internal Hackathon:** Secured **1st place** in the hardware domain at the college-level internal hackathon.
- **IIC Business Plan Competition:** Secured **2nd place** with *FuelTure*, a sustainable plastic-waste-to-fuel innovation.
- **HACKFEST 2025 (SAP):** Advanced beyond the college-level internal round with *FuelTure* to the regional presentation/selection stage.
- **IEEE R10 Innovation Challenge 2025:** Participated in the selection round with *FuelTure*, a plastic-waste valorization project.
- **OPTIMA Tech Fest 2025, IIT Kharagpur:** Secured **9th rank** while presenting *FuelTure* in the innovation and industrial-design showcase.
- **MCA Department Poster Competition, TINT 2025:** Secured **2nd Runner-Up** for presenting *FuelTure* and *Guardian Helm*.
- **Amity Innovathon 2025:** Participated in smart-wearable technology innovation and business pitching.
- **Incubes 2025, Techno International New Town:** Secured **5th rank at the college level** with *FuelTure* and was selected for mentoring by business professionals Arnab Ray and Arnab Basu.
- **Spice'24, Techno International New Town:** Presented the *Trackless Autonomous Tram* project.
- **Shark Tank Xavrang'25:** Presented the *Fire Extinguisher Car*, a sensor-based robotic fire-response prototype.
- **AURA'25:** Presented the *Fire Extinguisher Car* project.
- **Yukti Innovation Challenge 2025:** Advanced to the **second-round pitch** stage.
- **MSME Idea Hackathon 5.0:** Participated with a sustainable-technology innovation proposal; reported receiving ***[INR]*5 lakh in funding** through the central MSME initiative.
- **MSME Idea Hackathon 6.0, Ministry of MSME:** Submitted multiple innovation proposals through the official MSME Idea Hackathon platform.
- **The Lost Paper:** Secured **3rd place** in a story-writing competition.

Academic, Research and Innovation Participation

- **8th Regional Science & Technology Congress, WBDST:** Participated in the *Emerging Science & Technologies* programme.
- **SmartMed Bootcamp 2.0:** Participated in the programme on *Biomedical Signal Processing for Smart Wearables*.
- **IDE Bootcamp 2026:** Participated in the **5-Day IDE Bootcamp** programme.
- **International Conference – ETCSTSI 2026:** Participated in *Emerging Trends in Chemical Sciences Towards Sustainability & Interdisciplinarity*.
- **Asian Research Association Conference 2026:** Participated in the conference on *Advanced Materials: Bridging Physics, Chemistry, Biosciences and Computational Modelling*.
- **4th International Conference on Indian MSMEs @2047:** Participated at St. Xavier's University, Kolkata, March 2026.
- **Innovation to Protection – IPR Workshop:** Participated in the workshop and *MedDrip* project showcase.
- **Prabuddha 2026 Techfest – Science-to-Application Yatra:** Presented/participated with projects on *Parthenium to Biofuel Cell* and *Sensing Layers & Conductive Ink Development*.
- **UAS & Technology Bootcamp, NIT Durgapur:** Participated in workshops covering UAV design, MATLAB, SimScale, Mission Planner and TinyML.
- **IEEE STEM Summit 2025:** Participated as an attendee.
- **IEEE IMS Webinar Series:** Participated in sessions covering condition monitoring, wearable sensors, AI and signal processing.
- **MCA Inter-College Project/Poster Presentation Competitions:** Participated in multiple inter-college project and poster presentation events at Techno International New Town.

Awards and Achievements

- **1st Place – SIH 2026 Internal Hackathon, Techno International New Town:** Ranked **1st among 104 participating teams** in the institutional internal selection round for Smart India Hackathon 2026.
- **Grand Finalist – Smart India Hackathon (SIH) 2025, Hardware Edition:** Selected as a finalist with **Team ECOCELLUPAK_T16**.
- **3rd Place – SIH 2025 Internal Round:** Secured **3rd position** with Team **ECOCELLUPAK_T16** in the institutional selection round.
- **2nd Place – IIC Business Plan Competition:** Secured **2nd place** for *FuelTure*, a sustainable plastic-waste-to-fuel project.
- **2nd Runner-Up – IEEE Poster Competition 2025:** Secured **2nd Runner-Up** for presenting *FuelTure* and *Guardian Helm*.
- **Selected for State Round – HACKFEST 2025:** *FuelTure* selected for the State Round through the SAP-supported HACKFEST 2025.
- **Rs. 5 Lakh Grant – MSME Idea Hackathon 5.0:** Received a **Rs. 5 lakh grant** under the MSME Idea Hackathon 5.0 for innovative project development.
- **Best Paper Award – 8th Regional Science & Technology Congress, WBDST:** Received the **Best Paper Award** under the Emerging Science & Technology track.
- **IEEE R10 UPSKILL Kolkata 2025:** Won the **Innovate4Impact (PuranMed)** challenge.
- **Shortlisted – DDWS Innovation Challenge, IIT Madras:** Shortlisted for **Track 2 – Sustainable Plastic Waste Disposal System** and qualified for the **Ministry of Jal Shakti Masterclass**.
- **Selected for Mentorship – TINT Business Incubation Cell:** Selected for mentorship through the institutional business incubation programme.
- **3rd Place – Story Writing Competition:** Secured **3rd place** in "The Lost Paper."
- **Paper Shortlisted for Publication – ICMBP-2025:** Research paper on *Trackless Autonomous Trams* shortlisted for publication following presentation at ICMBP-2025.
- **Paper Selected for Journal Publication – CIACON 2025:** Research work on *Federated Learning for Collaborative Ageing and State of Health Prediction in E-Rickshaws* selected for journal publication.

Workshops, Webinars, Courses and Certifications

- **Arduino for Beginners – freeCodeCamp.org:** Completed an 8-hour introductory course on Arduino programming and embedded-system fundamentals.
- **Build an Electric Vehicle: A Beginner's Course – Udemy:** Completed an 8-hour course covering fundamental concepts of electric vehicle design and technology.
- **Front End Development – HTML:** Completed an 8-hour course covering HTML and fundamental front-end development concepts.
- **Machine Learning Training – Acmegrade:** Completed a structured machine-learning training programme.
- **VLSI:** Completed an extended VLSI course during September–November 2025.
- **IoT:** Completed an extended Internet of Things course during September–November 2025.
- **3D Printing and Additive Manufacturing Technology:** Completed a 16-hour course covering 3D printing and additive manufacturing technologies.
- **Autonomous Vehicles: A Complete Guide on Driverless Cars – Udemy:** Completed an 8-hour course covering autonomous and driverless vehicle technologies.
- **NPTEL – Ethics in Engineering Practice:** Completed the NPTEL course and secured **55%**.
- **NPTEL – Developing Soft Skills and Personality:** Completed the NPTEL course and secured **53%**.
- **Soft Skills for Business – TutorialsPoint:** Completed an 8-hour course focused on professional communication, business skills and workplace competencies.
- **IEEE EMBS Student Internship Program 2025:** Completed a one-month student internship programme from June 15 to July 15, 2025.
- **IEEE EMBS National Student Internship Program 2026:** Completed the one-month IEEE EMBS Pune Chapter National Student Internship Program from June 1 to June 30, 2026.
- **17th Bootcamp on Unmanned Aerial Systems and Related Technology:** Completed a **32-hour** technical bootcamp covering unmanned aerial systems and related technologies.
- **Instrumentation and Signal Processing Techniques for Condition Monitoring Applications:** Completed an 8-hour programme focused on instrumentation, signal processing and condition-monitoring applications.
- **Ignite Bootcamp – Idea to Plan:** Completed a 16-hour innovation and entrepreneurship bootcamp focused on transforming ideas into structured plans.
- **Online Lecture Series – Trends on Sensors, Signal Processing, IoT on Instrumentation & Measurement Applications:** Completed an 8-hour technical lecture series covering sensors, signal processing, IoT, instrumentation and measurement.
- **IEEE Webinar Series:** Participated in sessions including Ask Me Ep. 1, WISE, Biometric Systems and IEEE Cognitive Insights.
- **IoT Meets AI:** Participated in a technical session exploring the integration of artificial intelligence with Internet of

Things systems.

- **Syntrix – When AI Meets IoT:** Participated in a technical session on AI-enabled IoT applications.
- **Deep Learning and LLMs:** Participated in a technical session covering deep learning and large language models.
- **LLMs and ML Forecasting Techniques:** Participated in a technical session on large language models and machine-learning-based forecasting.
- **Designing with AI:** Participated in a technical session on AI-assisted design.
- **Tech-Powered Growth:** Participated in a technology-focused professional development session.
- **Green IT:** Participated in a session on sustainable and environmentally responsible information technology.
- **Thermal Cutting:** Participated in a technical session on thermal cutting technologies.
- **Visualizing Battery Capacity, PID Controller and EV Engineering – Forage:** Completed virtual learning modules related to battery systems, PID control and electric-vehicle engineering.
- **AutoCAD – Isometric vs. 3D Drawing, Learn Delta:** Completed training on isometric and three-dimensional CAD drawing.

Leadership and Academic Activities

- Former Head Boy, Delhi World Public School, Kalyani.
- Member of student-led R&D and mentoring teams.
- Participated in technical presentations, research conferences, innovation challenges and business-plan competitions.
- Active involvement in scientific communication, idea pitching and technical outreach.

Languages

English	Professional Proficiency	**Bengali** – Native Proficiency	
Hindi	Professional Working Proficiency		

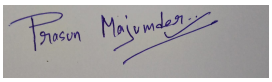
Research Interests

- Sustainable Materials and Waste Valorization
- Biomass Conversion and Biofuels
- Activated Carbon and Functional Carbon Materials
- Environmental Monitoring and Water Treatment
- Embedded Systems and IoT
- AI/ML and TinyML
- Intelligent Healthcare Systems
- Sensor Development and Signal Processing
- Energy Harvesting and Predictive Maintenance

Additional Information

Date of Birth	19 July 2005
Nationality	Indian
Location	Kalyani, Nadia, West Bengal, India
Hobbies	Technical Blogging, Story Writing, Public Speaking, DIY Electronics, AI/ML Experimentation and Startup Development

References and supporting documents can be provided upon request.



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